

Module 2 — Pythia8 Install & First Event

Supplementary to the sPHENIX Standalone Course

1. What Pythia8 provides

Pythia8 is a general-purpose event generator that simulates proton-proton, proton-electron, and e^+e^- collisions. It covers minimum-bias events, hard QCD, electroweak, heavy flavour, and BSM processes. The output is a list of final-state particles with four-momenta and colour connections.

2. Build steps

Download

```
cd $HOME/src
wget https://pythia.org/download/pythia83/pythia8312.tgz
tar xzf pythia8312.tgz
cd pythia8312
```

Configure & build

```
./configure --prefix=$HOME/opt/pythia8 --enable-shared
make -j$(nproc)
make install
```

Note: The build has no CMake — it uses a hand-written configure script. If configure fails because python3-config is missing, install python3-dev.

3. Optional extensions

Flag	What it adds
--with-hepmc2=DIR	Enables <code>Pythia8::HepMC2</code> output writer

--with-hepmc3=DIR	Enables HepMC3 output — newer format, preferred going forward
--with-lhapdf6=DIR	External PDF sets beyond the Pythia defaults
--with-fastjet3=DIR	Built-in jet clustering inside Pythia examples
--with-root=DIR	<code>main92.cc</code> -style ROOT tree writer example

For this course you only strictly need HepMC2 (for the capstone). The others are helpful but not required.

4. First event (main01.cc)

The main01.cc example ships with every Pythia release and is the fastest smoke test:

```
#include "Pythia8/Pythia8.h"
using namespace Pythia8;
int main() {
    Pythia pythia;
    pythia.readString("Beams:ecm = 200.");
    pythia.readString("HardQCD:all = on");
    pythia.readString("PhaseSpace:pTHatMin = 20.");
    pythia.init();
    for (int i=0; i<10; ++i) {
        if (!pythia.next()) continue;
        pythia.event.list();
    }
    pythia.stat();
    return 0;
}
```

Compile with:

```
g++ main01.cc -o main01 \
    -I$PYTHIA8DIR/include \
    -L$PYTHIA8DIR/lib -lpythia8 \
    -Wl, -rpath,$PYTHIA8DIR/lib
```

5. Anatomy of Pythia::next()

Each call to pythia.next() runs the full chain:

Phase	What happens
Hard process	Sample from chosen process group; sets pT-hat, partonic kinematics
ISR / FSR	Parton shower on incoming and outgoing partons
MPI	Multi-parton interactions underlying event
Hadronization	Lund string fragmentation

Decays	Unstable particles decay to stable final state
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6. Important global settings

```
Beams:eCM = 200.          // GeV, pp  $\sqrt{s}$ 
Tune:pp = 14              // Monash 2013 default for minimum bias
Random:setSeed = on
Random:seed = 0           // 0 = time-based
Next:numberShowEvent = 1 // print first event
Next:numberShowProcess = 0
```

7. Output patterns

Pythia's event is an STL-like container of `Pythia8::Particle`. You can loop directly:

```
for (int i=0; i<pythia.event.size(); ++i) {
    auto& p = pythia.event[i];
    if (!p.isFinal() || !p.isCharged()) continue;
    double pt = p.pT();
    double eta = p.eta();
}
```

8. Troubleshooting install

Symptom	Fix
undefined reference to `dlopen'	Add <code>-ldl</code> to link flags
libpythia8.so not found at runtime	Export <code>LD_LIBRARY_PATH</code> or use <code>-rpath</code>
HepMC writer class not found	You built Pythia before HepMC — rebuild with <code>--with-hepmc2</code>
configure: no acceptable C compiler	Install <code>build-essential</code> / <code>gcc</code>

9. What's in `share/Pythia8/xmldoc`?

That directory contains the master documentation for every Pythia setting. If a process or parameter name is new to you, grep the xmldoc to find its full description — it's better than any third-party reference.